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Java SE 27 & JDK 27

Class Writer

java.lang.Object
    java.io.Writer

All Implemented Interfaces:

Closeable, Flushable, Appendable, AutoCloseable

Direct Known Subclasses:

BufferedWriter, CharArrayWriter, FilterWriter, OutputStreamWriter, PipedWriter, PrintWriter, StringWriter

---

```
public abstract class Writer
extends Object
implements Appendable, Closeable, Flushable
```

Abstract class for writing to character streams. The only methods that a subclass must implement are write(char[], int, int), flush(), and close(). Most subclasses, however, will override some of the methods defined here in order to provide higher efficiency, additional functionality, or both.

Since:

1.1

See Also:

BufferedWriter, CharArrayWriter, FilterWriter, OutputStreamWriter, FileWriter, PipedWriter, PrintWriter, StringWriter, Reader

Field Summary

Fields

Modifier and Type	Field	Description
protected Object	**lock**	The object used to synchronize operations on this stream.

Constructor Summary

Constructors

Modifier	Constructor	Description
protected	**Writer**()	Creates a new character-stream writer whose critical sections will

	synchronize on the writer itself.
protected **Writer**(Object lock)	Creates a new character-stream writer whose critical sections will synchronize on the given object.

Method Summary

All Methods | Static Methods

Modifier and Type	Method	Description
Writer	**append**(char c)	Appends the specified character to this writer.
Writer	**append**(CharSequence csq)	Appends the specified character sequence to this writer.
Writer	**append**(CharSequence csq, int start, int end)	Appends a subsequence of the specified character sequence to this writer.
abstract void	**close**()	Closes the stream, flushing it first.
abstract void	**flush**()	Flushes the stream.
static Writer	**nullWriter**()	Returns a new Writer which discards all characters.
void	**write**(char[] cbuf)	Writes an array of characters.
abstract void	**write**(char[] cbuf, int off, int len)	Writes a portion of an array of characters.
void	**write**(int c)	

		Writes a single character.
void	write(String str)	Writes a string.
void	write(String str, int off, int len)	Writes a portion of a string.

Methods declared in class Object

clone, equals, finalize, getClass, hashCode, notify, notifyAll, toString, wait, wait, wait

Field Details

lock

```
protected Object lock
```

The object used to synchronize operations on this stream. For efficiency, a character-stream object may use an object other than itself to protect critical sections. A subclass should therefore use the object in this field rather than `this` or a synchronized method.

Constructor Details

Writer

```
protected Writer()
```

Creates a new character-stream writer whose critical sections will synchronize on the writer itself.

Writer

```
protected Writer(Object lock)
```

Creates a new character-stream writer whose critical sections will synchronize on the given object.

Parameters:

`lock` - Object to synchronize on

Method Details

nullWriter

```
public static Writer nullWriter()
```

Returns a new `Writer` which discards all characters. The returned stream is initially open. The stream is closed by calling the `close()` method. Subsequent calls to `close()` have no effect.

While the stream is open, the `append(char)`, `append(CharSequence)`, `append(CharSequence, int, int)`, `flush()`, `write(int)`, `write(char[])`, and `write(char[], int, int)` methods do nothing. After the stream has been closed, these methods all throw `IOException`.

The `object` used to synchronize operations on the returned `Writer` is not specified.

Returns:

a `Writer` which discards all characters

Since:

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write

```
public void write(int c)
           throws IOException
```

Writes a single character. The character to be written is contained in the 16 low-order bits of the given integer value; the 16 high-order bits are ignored.

Subclasses that intend to support efficient single-character output should override this method.

Parameters:

`c` - int specifying a character to be written

Throws:

`IOException` - If an I/O error occurs

write

```
public void write(char[] cbuf)
           throws IOException
```

Writes an array of characters.

Parameters:

`cbuf` - Array of characters to be written

Throws:

`IOException` - If an I/O error occurs

write

```
public abstract void write(char[] cbuf,
                           int off,
                           int len)
                    throws IOException
```

Writes a portion of an array of characters.

Parameters:

`cbuf` - Array of characters

`off` - Offset from which to start writing characters

`len` - Number of characters to write

Throws:

`IndexOutOfBoundsException` - Implementations should throw this exception if `off` is negative, or `len` is negative, or `off + len` is negative or greater than the length of the given array

`IOException` - If an I/O error occurs

write

```
public void write(String str)
           throws IOException
```

Writes a string.

Parameters:

`str` - String to be written

Throws:

`IOException` - If an I/O error occurs

write

```
public void write(String str,
                  int off,
                  int len)
           throws IOException
```

Writes a portion of a string.

Implementation Requirements:

The implementation in this class throws an `IndexOutOfBoundsException` for the indicated conditions; overriding methods may choose to do otherwise.

Parameters:

`str` - A String

`off` - Offset from which to start writing characters

`len` - Number of characters to write

Throws:

`IndexOutOfBoundsException` - Implementations should throw this exception if `off` is negative, or `len` is negative, or `off + len` is negative or greater than the length of the given string

`IOException` - If an I/O error occurs

append

```
public Writer append(CharSequence csq)
              throws IOException
```

Appends the specified character sequence to this writer.

An invocation of this method of the form `out.append(csq)` when `csq` is not `null`, behaves in exactly the same way as the invocation

```
    out.write(csq.toString())
```

Depending on the specification of `toString` for the character sequence `csq`, the entire sequence may not be appended. For instance, invoking the `toString` method of a character buffer will return a subsequence whose content depends upon the buffer's position and limit.

Specified by:

`append` in interface `Appendable`

Parameters:

`csq` - The character sequence to append. If `csq` is `null`, then the four characters `"null"` are appended to this writer.

Returns:

This writer

Throws:

`IOException` - If an I/O error occurs

Since:

1.5

append

```
public Writer append(CharSequence csq,
                     int start,
                     int end)
              throws IOException
```

Appends a subsequence of the specified character sequence to this writer.

An invocation of this method of the form `out.append(csq, start, end)` when `csq` is not `null` behaves in exactly the same way as the invocation

```
    out.write(csq.subSequence(start, end).toString())
```

Specified by:

`append` in interface `Appendable`

Parameters:

`csq` - The character sequence from which a subsequence will be appended. If `csq` is `null`, then characters will be appended as if `csq` contained the four characters `"null"`.

`start` - The index of the first character in the subsequence

`end` - The index of the character following the last character in the subsequence

Returns:

This writer

Throws:

`IndexOutOfBoundsException` - If `start` or `end` are negative, `start` is greater than `end`, or `end` is greater than `csq.length()`

`IOException` - If an I/O error occurs

Since:

1.5

append

```
public Writer append(char c)
              throws IOException
```

Appends the specified character to this writer.

An invocation of this method of the form `out.append(c)` behaves in exactly the same way as the invocation

```
    out.write(c)
```

Specified by:

append in interface Appendable

Parameters:

c - The 16-bit character to append

Returns:

This writer

Throws:

IOException - If an I/O error occurs

Since:

1.5

flush

```
public abstract void flush()
                    throws IOException
```

Flushes the stream. If the stream has saved any characters from the various write() methods in a buffer, write them immediately to their intended destination. Then, if that destination is another character or byte stream, flush it. Thus one flush() invocation will flush all the buffers in a chain of Writers and OutputStreams.

If the intended destination of this stream is an abstraction provided by the underlying operating system, for example a file, then flushing the stream guarantees only that bytes previously written to the stream are passed to the operating system for writing; it does not guarantee that they are actually written to a physical device such as a disk drive.

Specified by:

flush in interface Flushable

Throws:

IOException - If an I/O error occurs

close

```
public abstract void close()
                    throws IOException
```

Closes the stream, flushing it first. Once the stream has been closed, further write() or flush() invocations will cause an IOException to be thrown. Closing a previously closed stream has no effect.

Specified by:

`close` in interface `AutoCloseable`

Specified by:

`close` in interface `Closeable`

Throws:

`IOException` - If an I/O error occurs

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